

|                                                  | 0                                                                                                                                                                                                                 | 1 | 2 | 3                                 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
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| A                                                | E-01 COVER, METADATA, REVISION HISTORY                                                                                                                                                                            |   |   |                                   |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                | Compact Automated Two-Nozzle Bottle-Filling Cell                                                                                                                                                                  |   |   |                                   |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                | Electrical schematic set   Revision A   24 folios                                                                                                                                                                 |   |   |                                   |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                                                                                                   |   |   |                                   |   |   |   |   |   |   |    |                         |    |    |    |
|                                                  | CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES REVIEW, RISK ASSESSMENT, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY OR REGULATORY COMPLIANCE IS CLAIMED. |   |   |                                   |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                | 400/230 VAC 3P+N+PE, 50 Hz, TN-S assumed   24 VDC PELV   Indoor IP54 concept                                                                                                                                      |   |   |                                   |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                                                                                                   |   |   |                                   |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                   |   |   | Cover, metadata, revision history |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                   |   |   |                                   |   |   |   |   |   |   |    | Folio : 1/24            |    |    |    |

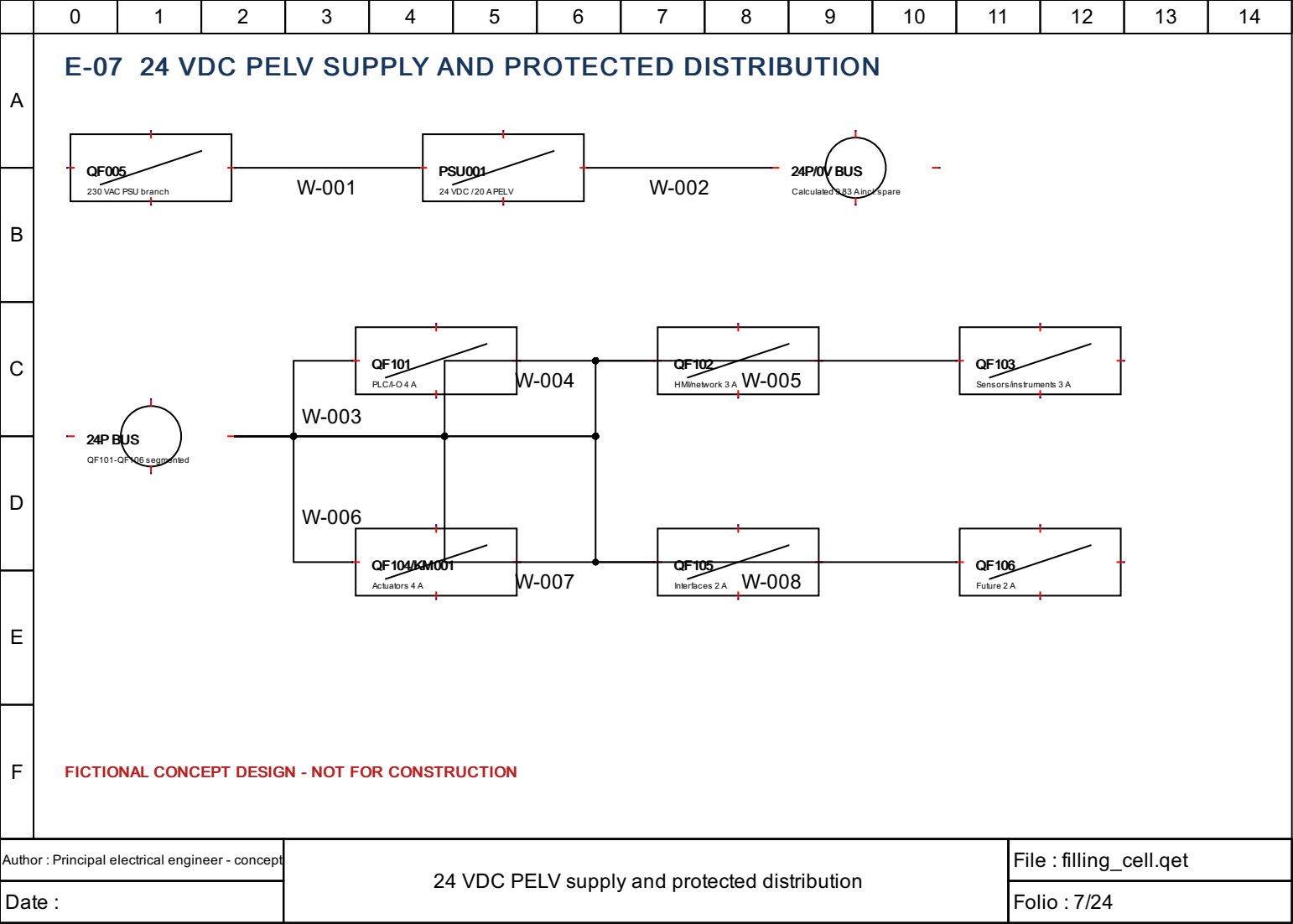
|                                                 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| A                                               | <div>E-02 DRAWING INDEX</div> <div><div>E-01 Cover, metadata, revision history</div><div>E-13 PLC digital outputs - module DO001</div><div>E-02 Drawing index</div><div>E-14 PLC digital outputs - module DO002 and spares</div><div>E-03 Design basis, legends, reference designations</div><div>E-15 PLC analog inputs and outputs</div><div>E-04 Functional overview and operating-state block diagram</div><div>E-16 Conveyor VFD control and power wiring</div><div>E-05 Single-line power distribution</div><div>E-17 Conveyor motor, PE and shield termination</div><div>E-06 Incoming isolation and branch protection</div><div>E-18 Dosing-pump VFD and motor interface</div><div>E-07 24 VDC PELV supply and protected distribution</div><div>E-19 Two-channel filling valves and flow instrumentation</div><div>E-08 Conceptual safety architecture</div><div>E-20 Bottle, gate, clamp, air and product sensors</div><div>E-09 PLC, HMI, Ethernet and controller power</div><div>E-21 HMI, stack light and capping-station interface</div><div>E-10 PLC digital inputs - module DI001</div><div>E-22 Terminal-strip plan and field connections</div><div>E-11 PLC digital inputs - module DI002</div><div>E-23 Cable overview, EMC and segregation notes</div><div>E-12 PLC digital inputs - module DI003 and spares</div><div>E-24 Spare capacity, general notes and commissioning prerequisites</div></div> |   |   |   |   |   |   |   |   |   |    |    |    |    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| B                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |    |    |    |    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |    |    |    |    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| A                                                | <div>E-03 DESIGN BASIS, LEGENDS, REFERENCE DESIGNATIONS</div> <div>REFERENCE DESIGNATION: =FC01+LOCATION-TAG   CP01 panel   FD01 field   OP01 operator   IF01 capping</div> <div>SUPPLY BASIS: 400/230 VAC, 3P+N+PE, 50 Hz, TN-S assumed. CONTROL: 24 VDC PELV.</div> <div>PNP ordinary sensors. Analog: isolated 4-20 mA; provisional raw 5530-27648. Field devices terminate at identified X strips.</div> <div>PROTECTION, CONDUCTOR SIZING, VOLTAGE DROP, FAULT WITHSTAND AND THERMAL RESULTS ARE PRELIMINARY.</div> <div>Wire IDs are unique physical segments. Cable/core and terminal references appear beside signal blocks; schedules are authoritative.</div> <div>No PL, SIL, category, CE, UL, SCCR or regulatory compliance is claimed.</div> <div>FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION</div> <div>CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES REVIEW, RISK ASSESSMENT, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY OR REGULATORY COMPLIANCE IS CLAIMED.</div> |   |   |                                               |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |                                               |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |                                               |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |                                               |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |                                               |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |                                               |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   | Design basis, legends, reference designations |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |                                               |   |   |   |   |   |   |    | Folio : 3/24            |    |    |    |

|                                                  | 0                                                          | 1 | 2 | 3                                                     | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
|--------------------------------------------------|------------------------------------------------------------|---|---|-------------------------------------------------------|---|---|---|---|---|---|----|-------------------------|----|----|----|
| A                                                | E-04 FUNCTIONAL OVERVIEW AND OPERATING-STATE BLOCK DIAGRAM |   |   |                                                       |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                |                                                            |   |   |                                                       |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                |                                                            |   |   |                                                       |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                |                                                            |   |   |                                                       |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                |                                                            |   |   |                                                       |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION            |   |   |                                                       |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                            |   |   | Functional overview and operating-state block diagram |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
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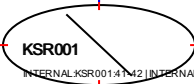
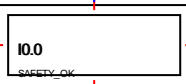
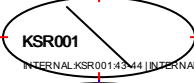
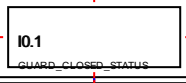
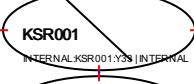
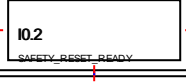
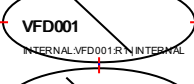
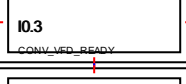
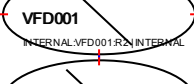
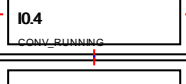
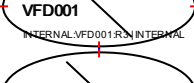
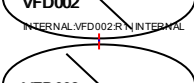
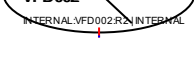
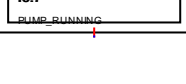
|                                                  | 0                                                                                                                                                                                                                                                                                                    | 1 | 2 | 3                                        | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
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| A                                                | E-06 INCOMING ISOLATION AND BRANCH PROTECTION                                                                                                                                                                                                                                                        |   |   |                                          |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                | <div><div><div>X1</div><div>L1L2L3NPE</div></div><div>W-001</div><div><div>QS001</div><div>Lockable 4-pole isolation</div></div><div>W-002</div><div><div>QF001</div><div>Main protection preliminary</div></div><div>W-003</div><div><div>3P+N+PE BUS</div><div>Branches E-06E-07</div></div></div> |   |   |                                          |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                |                                                                                                                                                                                                                                                                                                      |   |   |                                          |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                | <div><div><div>QF002</div><div>Conveyor branch 10 A</div></div><div>W-004</div><div><div>QF003</div><div>Pump branch 10 A</div></div><div>W-005</div><div><div>QF004</div><div>Fan branch 2 A</div></div><div>W-006</div><div><div>QF005</div><div>PSU branch 4 A 2P</div></div></div>               |   |   |                                          |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                | Breaking capacity, coordination, conductor ampacity and upstream protection require confirmed site data.                                                                                                                                                                                             |   |   |                                          |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                                                                                                                                                                                      |   |   |                                          |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                                      |   |   | Incoming isolation and branch protection |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                                                                                                      |   |   |                                          |   |   |   |   |   |   |    | Folio : 6/24            |    |    |    |



|                                                  | 0                                                                                                                                                                                                                                                                                       | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9                              | 10 | 11 | 12 | 13 | 14 |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|--------------------------------|----|----|----|----|----|
| A                                                | E-08 CONCEPTUAL SAFETY ARCHITECTURE                                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |   |                                |    |    |    |    |    |
| B                                                |                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |                                |    |    |    |    |    |
| C                                                |                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |                                |    |    |    |    |    |
| D                                                |                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |                                |    |    |    |    |    |
| E                                                | Standard PLC monitors status only. Restoration never starts motion. Required functions/performance are not determined here.                                                                                                                                                             |   |   |   |   |   |   |   |   |                                |    |    |    |    |    |
| F                                                | <div>FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION</div> <div>CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES REVIEW, RISK ASSESSMENT, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY OR REGULATORY COMPLIANCE IS CLAIMED.</div> |   |   |   |   |   |   |   |   |                                |    |    |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   | Conceptual safety architecture |    |    |    |    |    |
| Date :                                           |                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   |                                |    |    |    |    |    |
|                                                  |                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   | File : filling_cell.get        |    |    |    |    |    |
|                                                  |                                                                                                                                                                                                                                                                                         |   |   |   |   |   |   |   |   | Folio : 8/24                   |    |    |    |    |    |



|                                                  | 0                                                                                                                                                                                                                                                                                                                                                                           | 1 | 2 | 3                                       | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----------------------------------------|---|---|---|---|---|---|----|-------------------------|----|----|----|
| A                                                | E-09 PLC, HMI, ETHERNET AND CONTROLLER POWER                                                                                                                                                                                                                                                                                                                                |   |   |                                         |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                | <pre> graph LR     QF101[QF101<br/>24 Vlogic] -- W-001 --&gt; PLC001[PLC001<br/>CPU + Ethernet]     PLC001 -- W-002 --&gt; Expansion[D/I/O/AI/AO<br/>Expansion module] </pre>                                                                                                                                                                                               |   |   |                                         |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                |                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                         |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                | <pre> graph TD     SW001[SW001<br/>5-port Industrial Ethernet] --- Bus     subgraph Bus         direction LR         W003[W-003] --- W004[W-004] --- W005[W-005] --- W006[W-006]     end     Bus --- A001[A001<br/>7-inch HMI]     Bus --- VFD001[VFD001<br/>Conveyor service]     Bus --- VFD002[VFD002<br/>Pump service]     Bus --- PLC001[PLC001<br/>Controller] </pre> |   |   |                                         |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                |                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                         |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                | <p>FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION</p>                                                                                                                                                                                                                                                                                                                      |   |   |                                         |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                                                                                                             |   |   | PLC, HMI, Ethernet and controller power |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                         |   |   |   |   |   |   |    | Folio : 9/24            |    |    |    |

|                                                  |  | 0                                                                                                                  | 1     | 2                                                                                                                | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|--------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|----|----|----|----|----|
| E-10 PLC DIGITAL INPUTS - MODULE DI001           |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
| A                                                |  | <br>INTERNAL.KSR001.4 M2 INTERNAL  | W-001 | <br>I0.0<br>SAFETY_OK           |   |   |   |   |   |   |   |    |    |    |    |    |
| B                                                |  | <br>INTERNAL.KSR001.43 M4 INTERNAL | W-002 | <br>I0.1<br>GUARD_CLOSED_STATUS |   |   |   |   |   |   |   |    |    |    |    |    |
|                                                  |  | <br>INTERNAL.KSR001.Y38 INTERNAL   | W-003 | <br>I0.2<br>SAFETY_RESET_READY  |   |   |   |   |   |   |   |    |    |    |    |    |
| C                                                |  | <br>INTERNAL.VFD001.R INTERNAL     | W-004 | <br>I0.3<br>CONV_VFD_READY      |   |   |   |   |   |   |   |    |    |    |    |    |
|                                                  |  | <br>INTERNAL.VFD001.R2 INTERNAL    | W-005 | <br>I0.4<br>CONV_RUNNING        |   |   |   |   |   |   |   |    |    |    |    |    |
| D                                                |  | <br>INTERNAL.VFD001.R3 INTERNAL    | W-006 | <br>I0.5<br>CONV_FAULT          |   |   |   |   |   |   |   |    |    |    |    |    |
|                                                  |  | <br>INTERNAL.VFD002.R INTERNAL     | W-007 | <br>I0.6<br>PUMP_VFD_READY      |   |   |   |   |   |   |   |    |    |    |    |    |
| E                                                |  | <br>INTERNAL.VFD002.R2 INTERNAL    | W-008 | <br>I0.7<br>PUMP_RUNNING        |   |   |   |   |   |   |   |    |    |    |    |    |
|                                                  |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
|                                                  |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
| F                                                |  | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
| Author : Principal electrical engineer - concept |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
| Date :                                           |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
| PLC digital inputs - module DI001                |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
| File : filling_cell.get                          |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |
| Folio : 10/24                                    |  |                                                                                                                    |       |                                                                                                                  |   |   |   |   |   |   |   |    |    |    |    |    |

|                                                  |  | 0                                                                                                                                | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8                                                                                                                                  | 9 | 10 | 11                      | 12 | 13 | 14 |
|--------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|------------------------------------------------------------------------------------------------------------------------------------|---|----|-------------------------|----|----|----|
| E-11 PLC DIGITAL INPUTS - MODULE DI002           |  |                                                                                                                                  |   |   |   |   |   |   |   |                                                                                                                                    |   |    |                         |    |    |    |
| A                                                |  | <div><div><div>FT101</div><div>X4-004 W111A</div></div><div>W-001</div><div><div>I2.0</div><div>FLOW_1_DIAG_OK</div></div></div> |   |   |   |   |   |   |   | <div><div><div>IF501</div><div>X4-003 W501B</div></div><div>W-009</div><div><div>I3.0</div><div>CAP_READY</div></div></div>        |   |    |                         |    |    |    |
| B                                                |  | <div><div><div>FT102</div><div>X4-014 W112A</div></div><div>W-002</div><div><div>I2.1</div><div>FLOW_2_DIAG_OK</div></div></div> |   |   |   |   |   |   |   | <div><div><div>IF501</div><div>X4-004 W501A</div></div><div>W-010</div><div><div>I3.1</div><div>CAP_BUSY</div></div></div>         |   |    |                         |    |    |    |
|                                                  |  | <div><div><div>B121</div><div>X4-021 W121A</div></div><div>W-003</div><div><div>I2.2</div><div>VALVE_1_CLOSED</div></div></div>  |   |   |   |   |   |   |   | <div><div><div>IF501</div><div>X4-005 W501B</div></div><div>W-011</div><div><div>I3.2</div><div>CAP_FAULT</div></div></div>        |   |    |                         |    |    |    |
| C                                                |  | <div><div><div>B122</div><div>X4-022 W122A</div></div><div>W-004</div><div><div>I2.3</div><div>VALVE_2_CLOSED</div></div></div>  |   |   |   |   |   |   |   | <div><div><div>B106</div><div>X4-016 W106A</div></div><div>W-012</div><div><div>I3.3</div><div>PRODUCT_SUPPLY_OK</div></div></div> |   |    |                         |    |    |    |
|                                                  |  | <div><div><div>B123</div><div>X4-023 W123A</div></div><div>W-005</div><div><div>I2.4</div><div>GATE_RETRACTED</div></div></div>  |   |   |   |   |   |   |   | <div><div><div>S003</div><div>X4-031 W003A</div></div><div>W-013</div><div><div>I3.4</div><div>START_PB</div></div></div>          |   |    |                         |    |    |    |
| D                                                |  | <div><div><div>B124</div><div>X4-024 W124A</div></div><div>W-006</div><div><div>I2.5</div><div>GATE_EXTENDED</div></div></div>   |   |   |   |   |   |   |   | <div><div><div>S004</div><div>X4-032 W003B</div></div><div>W-014</div><div><div>I3.5</div><div>STOP_PB</div></div></div>           |   |    |                         |    |    |    |
|                                                  |  | <div><div><div>B125</div><div>X4-025 W125A</div></div><div>W-007</div><div><div>I2.6</div><div>CLAMP_RELEASED</div></div></div>  |   |   |   |   |   |   |   | <div><div><div>S005</div><div>X4-033 W003B</div></div><div>W-015</div><div><div>I3.6</div><div>FAULT_RESET_PB</div></div></div>    |   |    |                         |    |    |    |
| E                                                |  | <div><div><div>B126</div><div>X4-026 W126A</div></div><div>W-008</div><div><div>I2.7</div><div>CLAMP_ENGAGED</div></div></div>   |   |   |   |   |   |   |   | <div><div><div>S006</div><div>X4-034 W003A</div></div><div>W-016</div><div><div>I3.7</div><div>MODE_AUTO</div></div></div>         |   |    |                         |    |    |    |
| F                                                |  | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                  |   |   |   |   |   |   |   |                                                                                                                                    |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |  | PLC digital inputs - module DI002                                                                                                |   |   |   |   |   |   |   |                                                                                                                                    |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |  |                                                                                                                                  |   |   |   |   |   |   |   |                                                                                                                                    |   |    | Folio : 11/24           |    |    |    |

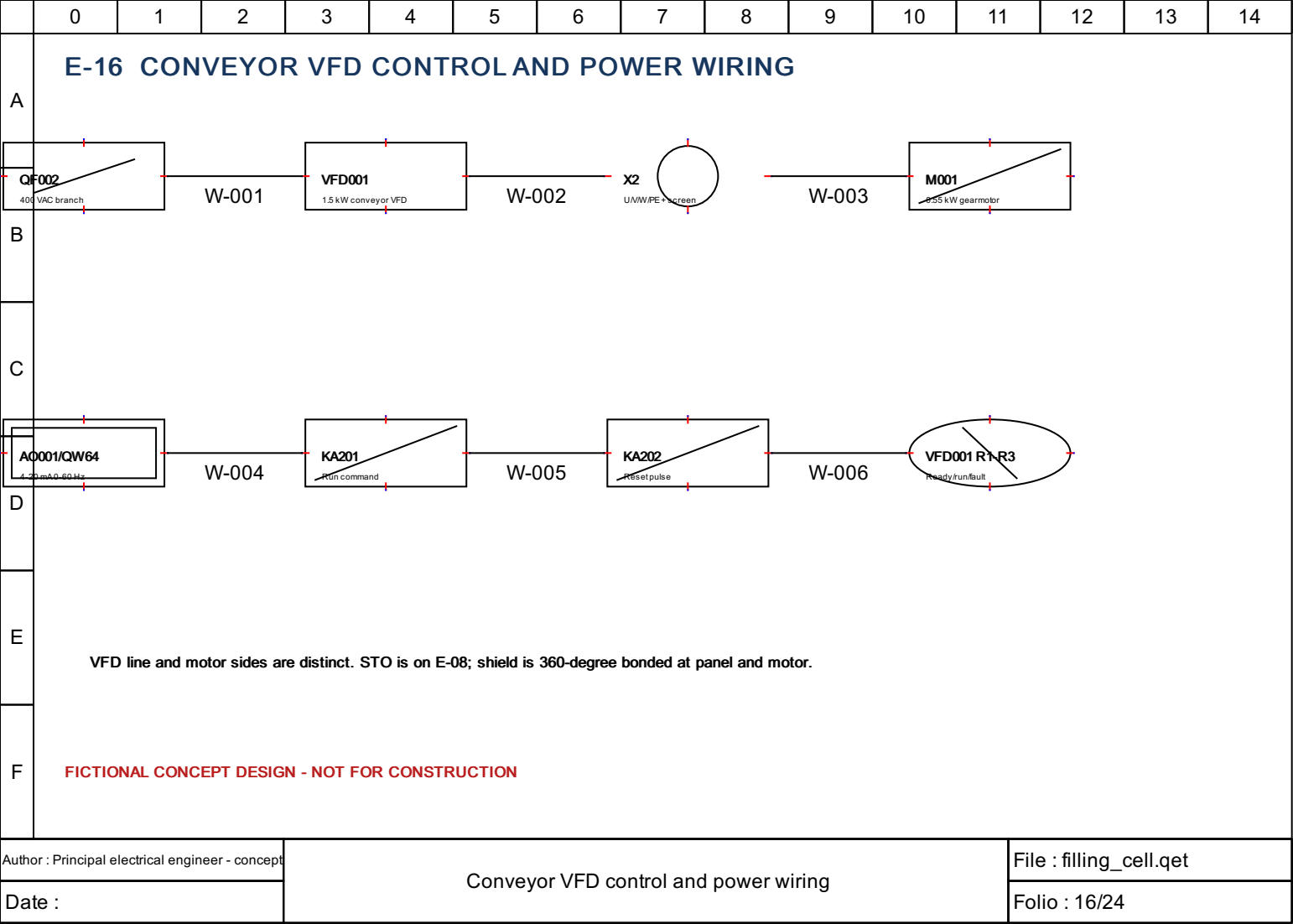
|                                                   | 0                                                                       | 1 | 2 | 3                                            | 4 | 5 | 6                                                    | 7 | 8 | 9                                                  | 10 | 11                      | 12    | 13 | 14 |                                               |  |  |
|---------------------------------------------------|-------------------------------------------------------------------------|---|---|----------------------------------------------|---|---|------------------------------------------------------|---|---|----------------------------------------------------|----|-------------------------|-------|----|----|-----------------------------------------------|--|--|
| E-12 PLC DIGITAL INPUTS - MODULE DI003 AND SPARES |                                                                         |   |   |                                              |   |   |                                                      |   |   |                                                    |    |                         |       |    |    |                                               |  |  |
| A                                                 | <div><div>S006</div><div>M1035 W0035</div></div>                        |   |   | W-001                                        |   |   | <div><div>I4.0</div><div>MODE_MANUAL</div></div>     |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-009 |    |    | <div><div>I5.0</div><div>SPARE_DI</div></div> |  |  |
| B                                                 | <div><div>VFD001</div><div>INTERNAL_VFD001_STO_MIN INTERNAL</div></div> |   |   | W-002                                        |   |   | <div><div>I4.1</div><div>CONV_STO_STATUS</div></div> |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-010 |    |    | <div><div>I5.1</div><div>SPARE_DI</div></div> |  |  |
|                                                   | <div><div>VFD002</div><div>INTERNAL_VFD002_STO_MIN INTERNAL</div></div> |   |   | W-003                                        |   |   | <div><div>I4.2</div><div>PUMP_STO_STATUS</div></div> |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-011 |    |    | <div><div>I5.2</div><div>SPARE_DI</div></div> |  |  |
| C                                                 | <div><div>RESERVED</div><div>NONE NONE</div></div>                      |   |   | W-004                                        |   |   | <div><div>I4.3</div><div>SPARE_DI</div></div>        |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-012 |    |    | <div><div>I5.3</div><div>SPARE_DI</div></div> |  |  |
|                                                   | <div><div>RESERVED</div><div>NONE NONE</div></div>                      |   |   | W-005                                        |   |   | <div><div>I4.4</div><div>SPARE_DI</div></div>        |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-013 |    |    | <div><div>I5.4</div><div>SPARE_DI</div></div> |  |  |
| D                                                 | <div><div>RESERVED</div><div>NONE NONE</div></div>                      |   |   | W-006                                        |   |   | <div><div>I4.5</div><div>SPARE_DI</div></div>        |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-014 |    |    | <div><div>I5.5</div><div>SPARE_DI</div></div> |  |  |
|                                                   | <div><div>RESERVED</div><div>NONE NONE</div></div>                      |   |   | W-007                                        |   |   | <div><div>I4.6</div><div>SPARE_DI</div></div>        |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-015 |    |    | <div><div>I5.6</div><div>SPARE_DI</div></div> |  |  |
| E                                                 | <div><div>RESERVED</div><div>NONE NONE</div></div>                      |   |   | W-008                                        |   |   | <div><div>I4.7</div><div>SPARE_DI</div></div>        |   |   | <div><div>RESERVED</div><div>NONE NONE</div></div> |    |                         | W-016 |    |    | <div><div>I5.7</div><div>SPARE_DI</div></div> |  |  |
| F                                                 | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                         |   |   |                                              |   |   |                                                      |   |   |                                                    |    |                         |       |    |    |                                               |  |  |
| Author : Principal electrical engineer - concept  |                                                                         |   |   | PLC digital inputs - module DI003 and spares |   |   |                                                      |   |   |                                                    |    | File : filling_cell.get |       |    |    |                                               |  |  |
| Date :                                            |                                                                         |   |   |                                              |   |   |                                                      |   |   |                                                    |    | Folio : 12/24           |       |    |    |                                               |  |  |

|                                                  |  | 0                                                        | 1     | 2                                         | 3 | 4 | 5 | 6 | 7                                           | 8     | 9                                           | 10 | 11                      | 12 | 13 | 14 |
|--------------------------------------------------|--|----------------------------------------------------------|-------|-------------------------------------------|---|---|---|---|---------------------------------------------|-------|---------------------------------------------|----|-------------------------|----|----|----|
| E-13 PLC DIGITAL OUTPUTS - MODULE DO001          |  |                                                          |       |                                           |   |   |   |   |                                             |       |                                             |    |                         |    |    |    |
| A                                                |  | <div>KA201</div> <div>INTERNAL_VFD001/DI1/INTERNAL</div> | W-001 | <div>Q0.0</div> <div>PUMP_RUN_CMD</div>   |   |   |   |   | <div>KA209</div> <div>X16-001/1W211/2</div> | W-009 | <div>Q1.0</div> <div>STACK_GREEN</div>      |    |                         |    |    |    |
| B                                                |  | <div>KA202</div> <div>INTERNAL_VFD001/DI2/INTERNAL</div> | W-002 | <div>Q0.1</div> <div>PUMP_RESET_CMD</div> |   |   |   |   | <div>KA210</div> <div>X16-002/1W211/2</div> | W-010 | <div>Q1.1</div> <div>STACK_AMBER</div>      |    |                         |    |    |    |
|                                                  |  | <div>KA203</div> <div>INTERNAL_VFD002/DI1/INTERNAL</div> | W-003 | <div>Q0.2</div> <div>PUMP_RUN_CMD</div>   |   |   |   |   | <div>KA211</div> <div>X16-003/1W211/2</div> | W-011 | <div>Q1.2</div> <div>STACK_RED</div>        |    |                         |    |    |    |
| C                                                |  | <div>KA204</div> <div>INTERNAL_VFD003/DI2/INTERNAL</div> | W-004 | <div>Q0.3</div> <div>PUMP_RESET_CMD</div> |   |   |   |   | <div>KA212</div> <div>X16-004/1W211/2</div> | W-012 | <div>Q1.3</div> <div>SOUNDER</div>          |    |                         |    |    |    |
|                                                  |  | <div>KA205</div> <div>X12-001/1W201/2</div>              | W-005 | <div>Q0.4</div> <div>PELL_VALVE_1</div>   |   |   |   |   | <div>KA213</div> <div>X16-001/1W501/2</div> | W-013 | <div>Q1.4</div> <div>PELL_READY</div>       |    |                         |    |    |    |
| D                                                |  | <div>KA206</div> <div>X12-003/1W202/2</div>              | W-006 | <div>Q0.5</div> <div>PELL_VALVE_2</div>   |   |   |   |   | <div>KA214</div> <div>X16-002/1W501/2</div> | W-014 | <div>Q1.5</div> <div>TRANSFER_REQUEST</div> |    |                         |    |    |    |
|                                                  |  | <div>KA207</div> <div>X12-005/1W203/2</div>              | W-007 | <div>Q0.6</div> <div>STOP_GATE</div>      |   |   |   |   | <div>KA215</div> <div>X16-006/1W501/2</div> | W-015 | <div>Q1.6</div> <div>PELL_FAULT</div>       |    |                         |    |    |    |
| E                                                |  | <div>KA208</div> <div>X12-007/1W204/2</div>              | W-008 | <div>Q0.7</div> <div>BOTTLE_CLAMP</div>   |   |   |   |   | <div>RESERVED</div> <div>NONE/NONE</div>    | W-016 | <div>Q1.7</div> <div>SPARE_DO</div>         |    |                         |    |    |    |
| F                                                |  | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION          |       |                                           |   |   |   |   |                                             |       |                                             |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |  | PLC digital outputs - module DO001                       |       |                                           |   |   |   |   |                                             |       |                                             |    | File : filling_cell.get |    |    |    |
| Date :                                           |  |                                                          |       |                                           |   |   |   |   |                                             |       |                                             |    | Folio : 13/24           |    |    |    |

|                                                    |                                                    | 0 | 1     | 2                                             | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12                      | 13 | 14 |                                                    |  |       |                                               |  |
|----------------------------------------------------|----------------------------------------------------|---|-------|-----------------------------------------------|---|---|---|---|---|---|---|----|----|-------------------------|----|----|----------------------------------------------------|--|-------|-----------------------------------------------|--|
| E-14 PLC DIGITAL OUTPUTS - MODULE DO002 AND SPARES |                                                    |   |       |                                               |   |   |   |   |   |   |   |    |    |                         |    |    |                                                    |  |       |                                               |  |
| A                                                  | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-001 | <div><div>Q2.0</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-009 | <div><div>Q3.0</div><div>SPARE_DO</div></div> |  |
| B                                                  | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-002 | <div><div>Q2.1</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-010 | <div><div>Q3.1</div><div>SPARE_DO</div></div> |  |
|                                                    | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-003 | <div><div>Q2.2</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-011 | <div><div>Q3.2</div><div>SPARE_DO</div></div> |  |
| C                                                  | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-004 | <div><div>Q2.3</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-012 | <div><div>Q3.3</div><div>SPARE_DO</div></div> |  |
|                                                    | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-005 | <div><div>Q2.4</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-013 | <div><div>Q3.4</div><div>SPARE_DO</div></div> |  |
| D                                                  | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-006 | <div><div>Q2.5</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-014 | <div><div>Q3.5</div><div>SPARE_DO</div></div> |  |
|                                                    | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-007 | <div><div>Q2.6</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-015 | <div><div>Q3.6</div><div>SPARE_DO</div></div> |  |
| E                                                  | <div><div>RESERVED</div><div>NONE NONE</div></div> |   | W-008 | <div><div>Q2.7</div><div>SPARE_DO</div></div> |   |   |   |   |   |   |   |    |    |                         |    |    | <div><div>RESERVED</div><div>NONE NONE</div></div> |  | W-016 | <div><div>Q3.7</div><div>SPARE_DO</div></div> |  |
| F                                                  | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION    |   |       |                                               |   |   |   |   |   |   |   |    |    |                         |    |    |                                                    |  |       |                                               |  |
| Author : Principal electrical engineer - concept   |                                                    |   |       | PLC digital outputs - module DO002 and spares |   |   |   |   |   |   |   |    |    | File : filling_cell.get |    |    |                                                    |  |       |                                               |  |
| Date :                                             |                                                    |   |       |                                               |   |   |   |   |   |   |   |    |    | Folio : 14/24           |    |    |                                                    |  |       |                                               |  |

|                                    |                                                                                                                                                 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| E-15 PLC ANALOG INPUTS AND OUTPUTS |                                                                                                                                                 |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| A                                  | <div><div><div>FT101</div><div>AI4005.006 W111.5%</div></div><div>W-001</div><div><div>IW64</div><div>FLOW_1_RATE</div></div></div>             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| B                                  | <div><div><div>FT102</div><div>AI4015.016 W112.5%</div></div><div>W-002</div><div><div>IW66</div><div>FLOW_2_RATE</div></div></div>             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
|                                    | <div><div><div>VFD001</div><div>INTERNAL.VFD001.AIN INTERNAL</div></div><div>W-003</div><div><div>IW68</div><div>CONV_CURRENT</div></div></div> |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| C                                  | <div><div><div>VFD002</div><div>INTERNAL.VFD002.AIN INTERNAL</div></div><div>W-004</div><div><div>IW70</div><div>PUMP_CURRENT</div></div></div> |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| D                                  | <div><div><div>RESERVED</div><div>NONE NONE</div></div><div>W-005</div><div><div>IW72</div><div>SPARE_AI</div></div></div>                      |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
|                                    | <div><div><div>RESERVED</div><div>NONE NONE</div></div><div>W-006</div><div><div>IW74</div><div>SPARE_AI</div></div></div>                      |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
|                                    | <div><div><div>RESERVED</div><div>NONE NONE</div></div><div>W-007</div><div><div>IW76</div><div>SPARE_AI</div></div></div>                      |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| E                                  | <div><div><div>RESERVED</div><div>NONE NONE</div></div><div>W-008</div><div><div>IW78</div><div>SPARE_AI</div></div></div>                      |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| F                                  | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                                 |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |

|                                                  |  |                               |                         |
|--------------------------------------------------|--|-------------------------------|-------------------------|
| Author : Principal electrical engineer - concept |  | PLC analog inputs and outputs | File : filling_cell.get |
| Date :                                           |  |                               | Folio : 15/24           |

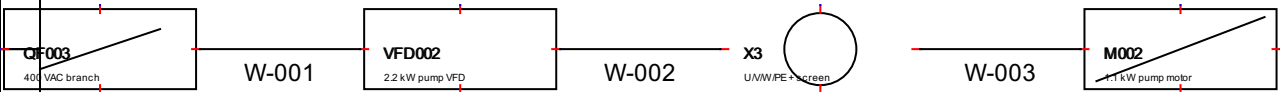




|                                                  | 0                                                                                                                                                                                                                                                                                                                                                    | 1 | 2 | 3                                         | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-------------------------------------------|---|---|---|---|---|---|----|-------------------------|----|----|----|
| A                                                | E-17 CONVEYOR MOTOR, PE AND SHIELD TERMINATION                                                                                                                                                                                                                                                                                                       |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                | <p>VFD001 U/V/W<br/>Motor output</p> <p>W-001</p> <p>X2-001-003<br/>Power terminals</p> <p>W-002</p> <p>W301<br/>4G1.5 shielded</p> <p>W-003</p> <p>M001<br/>U/V/W + PE</p> <p>PE001<br/>Panel PE bar</p> <p>W-004</p> <p>X2-PE1<br/>PE terminal</p> <p>W-005</p> <p>W301 PE/screen<br/>Both-end bond</p> <p>W-006</p> <p>M001 PE<br/>Motor bond</p> |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                |                                                                                                                                                                                                                                                                                                                                                      |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                |                                                                                                                                                                                                                                                                                                                                                      |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                |                                                                                                                                                                                                                                                                                                                                                      |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                                                                                                                                                                                                                                      |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                                                                                      |   |   | Conveyor motor, PE and shield termination |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                                                                                                                                                      |   |   |                                           |   |   |   |   |   |   |    | Folio : 17/24           |    |    |    |

E-18 DOSING-PUMP VFD AND MOTOR INTERFACE

A



B

C



D

E

Pump must not dead-head. Electrical stop logic is not a substitute for process relief/bypass assessment.

F

FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION

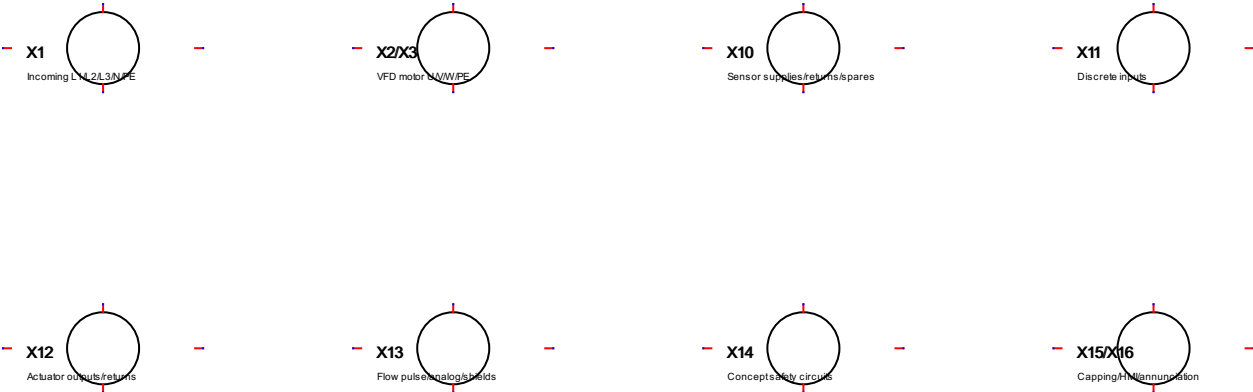
|                                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1 | 2 | 3                                                   | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----------------------------------------------------|---|---|---|---|---|---|----|-------------------------|----|----|----|
| A                                                | E-19 TWO-CHANNEL FILLING VALVES AND FLOW INSTRUMENTATION                                                                                                                                                                                                                                                                                                                                                                                               |   |   |                                                     |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |                                                     |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                | <div><div><div>M002/PUMP<br/>Common hygienic pump</div><div>MANIFOLD<br/>Two equal branches</div><div>W-001</div></div></div>                                                                                                                                                                                                                                                                                                                          |   |   |                                                     |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |                                                     |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                | <div><div><div>FT101<br/>Pulse + 4-20 mA</div><div>W-002</div><div>YV101<br/>Fail-closed valve 1</div><div>W-003</div><div>B121<br/>Closed feedback</div><div>W-004</div><div>BOTTLE 1<br/>0.50 L concept</div></div><div><div><div>FT102<br/>Pulse + 4-20 mA</div><div>W-005</div><div>YV102<br/>Fail-closed valve 2</div><div>W-006</div><div>B122<br/>Closed feedback</div><div>W-007</div><div>BOTTLE 2<br/>0.50 L concept</div></div></div></div> |   |   |                                                     |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |                                                     |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   | Two-channel filling valves and flow instrumentation |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |   |                                                     |   |   |   |   |   |   |    | Folio : 19/24           |    |    |    |



|                                                  |                                                                                                                                                                                                                     |   |   |                                                |   |   |   |   |   |   |    |                         |    |    |    |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|------------------------------------------------|---|---|---|---|---|---|----|-------------------------|----|----|----|
|                                                  | 0                                                                                                                                                                                                                   | 1 | 2 | 3                                              | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
| A                                                | E-21 HMI, STACK LIGHT AND CAPPING-STATION INTERFACE                                                                                                                                                                 |   |   |                                                |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                | <div><div><div>PLC001</div><div>Stack/light data</div></div><div>W-001</div><div><div>SW001</div><div>Ethernet</div></div><div>W-002</div><div><div>A001</div><div>HMI pages/access</div></div></div>               |   |   |                                                |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                | <div><div><div>KA209/KA212</div><div>Stack/light relays</div></div><div>W-003</div><div><div>H001/H004</div><div>R/AG + sounder</div></div></div>                                                                   |   |   |                                                |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                |                                                                                                                                                                                                                     |   |   |                                                |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                | <div><div><div>PLC/KA213-215</div><div>Cell output</div></div><div>W-004</div><div><div>X15W501</div><div>12-core handshake</div></div><div>W-005</div><div><div>IF501</div><div>Ready/busy/fault</div></div></div> |   |   |                                                |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                | FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION                                                                                                                                                                     |   |   |                                                |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                     |   |   | HMI, stack light and capping-station interface |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                     |   |   |                                                |   |   |   |   |   |   |    | Folio : 21/24           |    |    |    |

|   | 0                                              | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|------------------------------------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| A | E-22 TERMINAL-STRIP PLAN AND FIELD CONNECTIONS |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| B |                                                |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| C |                                                |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| D |                                                |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| E |                                                |   |   |   |   |   |   |   |   |   |    |    |    |    |    |
| F |                                                |   |   |   |   |   |   |   |   |   |    |    |    |    |    |

E-22 TERMINAL-STRIP PLAN AND FIELD CONNECTIONS



Canonical terminal plan contains 133 core/terminal records. Field and panel sides are explicit in the schedule.

FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION

|                                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 | 2 | 3                                         | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                      | 12 | 13 | 14 |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-------------------------------------------|---|---|---|---|---|---|----|-------------------------|----|----|----|
| A                                                | <div>E-23 CABLE OVERVIEW, EMC AND SEGREGATION NOTES</div> <div><div><div><div>W001-W003</div><div>Safety/operator control</div></div><div><div>W101-W126</div><div>PNP sensor cables</div></div><div><div>W111-W112</div><div>Shielded flow instrument</div></div><div><div>W201-W211</div><div>Actuators/annunciation</div></div><div><div>W301-W302</div><div>Shielded VFD motor</div></div><div><div>W401-W402</div><div>HMI/network/power</div></div><div><div>W501</div><div>Capping handshake</div></div></div><div>Segregate VFD outputs from 24 VDC and instrument wiring. Terminate shields exactly as scheduled; final routing/length/derating remain open.</div><div>FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION</div></div> |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| B                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| C                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| D                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| E                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| F                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                           |   |   |   |   |   |   |    |                         |    |    |    |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   | Cable overview, EMC and segregation notes |   |   |   |   |   |   |    | File : filling_cell.get |    |    |    |
| Date :                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |                                           |   |   |   |   |   |   |    | Folio : 23/24           |    |    |    |

|                                                  | 0                                                                                                                                                                                                                                                                                  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11                                                            | 12 | 13 | 14                      |  |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|----|---------------------------------------------------------------|----|----|-------------------------|--|
| A                                                | E-24 SPARE CAPACITY, GENERAL NOTES AND COMMISSIONING PREREQUISITES                                                                                                                                                                                                                 |   |   |   |   |   |   |   |   |   |    |                                                               |    |    |                         |  |
| B                                                | <div><div><div>DI</div><div>35/48 used/13 spare</div></div><div><div>DO</div><div>16/32 used/17 spare</div></div><div><div>AI</div><div>4/8 used/4 spare</div></div><div><div>AO</div><div>2/4 used/2 spare</div></div></div>                                                      |   |   |   |   |   |   |   |   |   |    |                                                               |    |    |                         |  |
| C                                                | <p>GENERAL: final device selections, fault-current/coordination, conductor sizes, voltage drop, EMC and thermal analysis require project data.</p>                                                                                                                                 |   |   |   |   |   |   |   |   |   |    |                                                               |    |    |                         |  |
| D                                                | <p>NETWORK: Ethernet loss does not substitute for hardwired stop; PLC controls drives through hardwired commands and analog references.</p> <p>COMMISSIONING: test every I/O point, scaling, sensor, valve, drive, handshake, restart and fault response before process use.</p>   |   |   |   |   |   |   |   |   |   |    |                                                               |    |    |                         |  |
| E                                                | <p>SAFETY: this drawing only describes a conceptual boundary. Qualified risk assessment, verification and validation remain mandatory.</p>                                                                                                                                         |   |   |   |   |   |   |   |   |   |    |                                                               |    |    |                         |  |
| F                                                | <p>FICTIONAL CONCEPT DESIGN - NOT FOR CONSTRUCTION</p> <p>CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES REVIEW, RISK ASSESSMENT, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY SAFETY ENGINEER.<br/>NO PERFORMANCE LEVEL, SIL CATEGORY OR REGULATORY COMPLIANCE IS CLAIMED.</p> |   |   |   |   |   |   |   |   |   |    |                                                               |    |    |                         |  |
| Author : Principal electrical engineer - concept |                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |    | Spare capacity, general notes and commissioning prerequisites |    |    | File : filling_cell.get |  |
| Date :                                           |                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |    |                                                               |    |    | Folio : 24/24           |  |